SHTPC: Stateful Hidden Trajectory-Payload Consistency for Private Heatmap Aggregation

Yingjie Wang, Hao Wang, Wenhan Hou, Haojun Teng, Kaile Su, Zhipeng Cai

***Abstract*—Managed heatmap services need to screen location reports before private aggregation without exposing coordinates or primary cells. Existing compositions can validate a hidden trajectory and an encoded payload separately, yet no single relation guarantees that both refer to the same primary cell. We introduce Stateful Hidden Trajectory-Payload Consistency (SHTPC), a stateful Groth16 admission relation. It binds accepted-predecessor continuity, window reachability, the coordinate-derived primary, and A/R payload commitments within one report context. The evaluated circuit has 3,547 constraints, 16 public inputs, and 25 private inputs, and it passes all 22 static checks. Across 15 N=1000 dataset-seed units, 225,000 proofs verify with 0 failures. Under one fixed three-seed configuration, SHTPC yields positive released-set Jaccard differences against a mechanism-level external ESA adapter on all five datasets, although two differences are small. SHTPC proves post-enrollment proof-to-payload consistency. It does not establish physical presence, eliminate metadata leakage, or reject false reports that remain inside the public mobility envelope.**

***Index Terms*—Mobile crowdsensing, Private aggregation, Zero-knowledge proofs, Admission consistency, Secure aggregation.**

I. Introduction

MANAGED private heatmap services turn enrolled-device reports into population-level spatial statistics for traffic engineering, public transit planning, and environmental monitoring [1, 2]. Such services can be viewed as a managed form of mobile crowdsensing, where enrollment and admission state are available but raw coordinates and primary heatmap cells should remain hidden. Such deployments often require three properties at once. A verifier should admit only reports that are consistent with a public mobility envelope. The verifier should not learn exact coordinates or primary heatmap cells. The aggregation roles should receive only encrypted or share-encoded payloads before the final differentially private release [3–8]. This setting is narrower than fully anonymous trajectory publication. It assumes enrollment, persistent pseudonymous admission state, and separated aggregation roles.

The structural gap is a proof-to-payload gap. A coordinate-hiding proof can show that x_t follows the accepted state and remains inside a public K-window envelope, while a separate payload passes encoding and context checks. If the two checks use independent private primary witnesses, a client can prove for $p_t = \mathrm{Grid}(x_t)$ but route a one-hot contribution to $p'_t \neq p_t$. Context hashes authenticate the frame, not this hidden equality; the Shuffler cannot see x_t, and an Aggregator cannot inspect the proof witness. Avoiding the mismatch therefore requires revealing the primary, trusting a plaintext checker, or proving coordinate-to-payload equality inside one relation. This paper studies the third option.

SHTPC addresses this gap as a stateful zero-knowledge admission layer placed before private heatmap aggregation. Each submission carries a compact SHTPCMain Groth16 proof that links the current hidden location to the previous accepted state, enforces per-step speed bounds and Anchor Distance Window Continuity (ADWC), derives the hidden primary heatmap cell, and binds that primary to report and A/R share-content commitments under the current report context. The Shuffler verifies proof validity, chain continuity, timing tokens, and transport authentication before forwarding accepted reports to secure aggregation. The downstream ε-DP heatmap release is unchanged [9].

The claim is intentionally limited to post-enrollment data-plane consistency. SHTPC does not prove physical presence, prevent false genesis at enrollment, prevent voluntary credential or device transfer, provide trajectory anonymity, secure a malicious Shuffler, or reject reports that remain fully inside the public envelope. The Shuffler also learns L_S, including chain topology, timing, mode or tier metadata, and accept or reject outcomes. These signals are not covered by the public heatmap's $\varepsilon_{\text{total}}$-DP accounting. The security contribution is therefore the SHTPC admission-consistency property itself. Groth16, commitments, ADWC, share digests, and DP release are building blocks for realizing that property under explicit leakage and trust boundaries.

This paper makes three contributions.

- Formalizes SHTPC as a stateful post-enrollment admission-consistency property for coordinate-hiding heatmap aggregation, with explicit predecessor, K-window, hidden-primary, payload-binding, leakage, and corruption boundaries.
- Designs a zero-knowledge admission layer that binds these predicates in one accepted statement. The paper further gives separation arguments showing why stateless

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ZK, VDAF-style validation, context commitments, and post-admission defenses do not imply SHTPC, and gives an end-to-end payload-consistency theorem connecting the circuit witness, public binders, ciphertext blobs, Aggregator validation, and the final clipped one-hot contribution.
- Evaluates SHTPC as admission-boundary enforcement, with design-space baselines, measured proof-to-payload substitutions, state/window-boundary checks, metadata leakage analysis, utility cost, and reproducible artifact evidence under the stated deployment boundary. The fixed comparison has a positive observed Jaccard difference for every one of the 15 dataset-seed pairs, although two dataset means differ only slightly. The full-path matrix verifies 225,000 proofs with 0 failures.

II. Related Work

SHTPC is closest to work on private location analytics, aggregate-input validation, and proofs over hidden data. These systems protect outputs, measurements, or location claims, whereas SHTPC targets the consistency of an accepted hidden report: predecessor state, K-window reachability, coordinate-derived primary, and routed payload must agree before aggregation.

Prio/VDAF validates encoded measurements, and Prochlo/Nebula separates encode, shuffle, and analyze roles, but neither supplies accepted-predecessor, K-anchor, and hidden same-primary equality by itself [4, 7, 8, 10]. EIFFeL-style or stateless ZK filters can prove a report-local predicate, while robust aggregation acts after admission [11–15]. Proof-of-location and trusted filters strengthen physical or plaintext checks but change the target trust boundary [16]. The measured Stateful-ContextCommit baseline instead preserves hidden state and context yet omits in-relation same-primary equality; the full design-space comparison appears in Table II.

A. Private Location Analytics

Private location analytics primarily protects either the published statistic or the submitted location representation. Differential privacy, local/shuffle privacy, geo-indistinguishability, trajectory anonymization, and private trajectory synthesis reduce what is revealed by outputs or traces [3, 17–39]. These mechanisms are compatible with a heatmap release and can be used downstream of SHTPC. Their trust boundary is nevertheless output privacy or trace obfuscation, not admission consistency. They do not decide whether a hidden coordinate is reachable from the previously accepted hidden state or whether its primary cell is the one routed into aggregation.

B. Private Aggregation and Input Validation

Private aggregation and validation systems protect messages, shares, or encoded measurements after collection. Prochlo, Nebula, secure aggregation, Prio, Poplar, ACORN, and VDAF-style protocols separate collection roles and validate encoded measurements [4–10, 40–46]. Robust aggregation and federated-learning integrity defenses reduce the effect of malicious inputs after admission [11–15, 47–51]. These systems are essential once an input has been admitted, but their validation predicates are normally measurement-local or aggregate-robustness oriented. SHTPC addresses the preceding admission boundary. The hidden trajectory relation and heatmap payload must be one accepted statement before secure aggregation separates the report across roles.

C. Location Authenticity and ZK Binding

Location authenticity and private accounting systems provide useful deployment building blocks but target different predicates. GPS-spoofing defenses, GNSS authentication, anonymous credentials, trusted processors, and secure elements can strengthen enrollment or possession claims [16, 52–58]. VPriv, PrETP, PriPAYD, and private tolling protocols combine hidden location evidence with charging or audit rules [59–65]. These lines of work can improve how a client is enrolled, how a device anchors its sensor evidence, or how a private service audits billing outcomes. Their predicates do not express repeated heatmap admission with predecessor continuity, K-window reachability, and same-primary payload binding.

Zero-knowledge proof systems provide the substrate for checking private relations without revealing witnesses [66–80]. For SHTPC, the central question is not merely proof size or verifier time, but which equalities are enforced inside one relation. Prior analyses show that missing constraints can create practical ZK soundness bugs [81]. Accordingly, SHTPC's C17–C21 constraints derive the hidden primary from the hidden coordinate and bind that primary to the report-level payload and A/R share commitments under one report context.

III. Problem Definition and Composition Gaps

SHTPC decides whether a hidden report may enter private heatmap aggregation. It targets post-enrollment consistency, not physical-presence proof or trajectory anonymity.

A. Report and Admission Model

Let uid be the pseudonymous chain handle, e the enrollment/reset epoch, t the admission round, w the release window, $x_t = (x, y)$ the hidden coordinate, and L_S the Shuffler-visible leakage.

The monitored region is partitioned into a public $W \times H$ grid with cell size s. A hidden coordinate determines

$$\mathrm{primary}_t = W\lfloor y_t/s \rfloor + \lfloor x_t/s \rfloor.$$

where W is the grid width in cells and $D = WH$ is the number of heatmap cells. The evaluated grid uses $s = 100\,\mathrm{m}$, $W = 100$, and $H = 100$. In each release window, an accepted client contributes a clipped one-hot vector $\tilde{\mathbf{u}}_w^{(i)} \in \mathbb{R}^D$ with $\|\tilde{\mathbf{u}}_w^{(i)}\|_1 \le 1$. The aggregate target is $\mathbf{s}_w = \sum_i \tilde{\mathbf{u}}_w^{(i)}$. SHTPC does not modify this target or the downstream pure $\varepsilon_{\mathrm{total}}$-DP heatmap release [3]. It only decides whether a hidden report is admissible before aggregation. This DP guarantee applies to the released heatmap only. It does not protect L_S metadata.

Salted commitments and a zero-knowledge admission relation check this hidden state without exposing coordinates or primary cells; Sections IV-C and IV-H give the concrete Groth16/Poseidon/MiMC7 instantiation [70, 78, 79].

B. Stateful Hidden Trajectory-Payload Consistency

The central property is stateful because the Shuffler stores the last accepted hidden state for each enrolled identity. It is hidden because the checked coordinate and primary cell remain circuit witnesses.

Definition 1 (Anchor Distance Window Continuity (ADWC) Predicate)**.** For trajectory $\{p_i\}_{i\geq 0}$ with $p_i = (x_i, y_i)$, window parameter $K \geq 1$, anchor $a_i = p_{\max(0,i-K)}$, tier speed τ_u, Shuffler-issued interval Δt_i, window time budget W_i, and deployment cap $\mathrm{cap}_{\mathrm{policy}}$, ADWC holds at step i iff

$$\begin{aligned} &\|p_i - p_{i-1}\|^2 \leq \tau_u^2 \Delta t_i^2, \\ &\|p_i - a_i\|^2 \leq \min\left(\tau_u^2 W_i^2, \mathrm{cap}_{\mathrm{policy}}^2\right). \end{aligned} \tag{1}$$

where $W_i = K\Delta t_{\mathrm{slot}}$ in the fixed-slot artifact evaluated in this paper, and a variable-slot deployment must instead use a Shuffler-signed window-level time sum. For $i < K$, $a_i = p_0$, so warmup uses the same circuit path rather than skipping anchor checks.

Definition 2 (Stateful Hidden Trajectory-Payload Consistency (SHTPC))**.** An accepted post-enrollment report satisfies SHTPC if five conditions hold for the same hidden report witness. First, the predecessor commitment opens to the previous accepted Shuffler state for the same uid and epoch. Second, the current hidden coordinate satisfies ADWC with respect to the predecessor and the K-window anchor. Third, the primary heatmap cell is derived from the hidden coordinate by the public grid rule. Fourth, the submitted aggregation payload is bound to that same hidden primary cell under the report context. Fifth, uid, epoch, round, timing token, and report context match the accepted public statement.

The SHTPCMAIN relation proves one witness for state, mobility, grid, payload, and context consistency:

$$\mathcal{R}_{\mathrm{SHTPC}} = R_{\mathrm{state}} \wedge R_{\mathrm{ADWC}} \wedge R_{\mathrm{grid}} \wedge R_{\mathrm{payload}} \wedge R_{\mathrm{context}}. \tag{2}$$

Here R_{state} opens the accepted predecessor and anchor; R_{ADWC} enforces step and window bounds; R_{grid} derives $\mathrm{primary}_t$ from x_t; R_{payload} binds that primary to the report and A/R share commitments; and R_{context} binds identity, epoch, round, timing token, and report context. The Shuffler checks public state, timing, and transport hashes; each Aggregator validates only its channel-local package.

C. Leakage Boundary and Goal Scope

Definition 3 (Explicit Shuffler Leakage Function)**.** The Shuffler-visible leakage L_S contains the uid or pseudonymous chain handle, epoch, round, timing slot, interarrival pattern, chain edge, mode or tier tag, proof outcome, accept or reject outcome, reset or warmup status, message-size class, and public context identifiers. It excludes raw coordinates and primary cells, but it includes enough topology and timing information to reveal behavioral patterns. SHTPC therefore proves coordinate-and-primary hiding only for equal-L_S worlds. It does not provide trajectory anonymity or report unlinkability.

TABLE I
CLAIM AND CORRUPTION BOUNDARY FOR SHTPC.

Claim or profile	Boundary
Raw coordinate hiding	Game 3, equal-L_S single-server views.
Primary hiding	Game 3, equal-L_S. Metadata remains visible.
HBC Shuffler	Sees L_S, including metadata that is not DP-protected.
One Aggregator	Sees one share package. Single-share privacy applies.
Trajectory anonymity	Not claimed.
Report unlinkability	Not claimed because per-uid state is required.
Output heatmap	Pure $\varepsilon_{\mathrm{total}}$-DP only for released statistics.
A+R collusion	Outside input-privacy claim. Output DP remains.
Malicious Shuffler	Outside claim. Requires audit or replicated logs.
False genesis/transfer	Deployment scope.
In-envelope false reports	Residual risk accepted by design.

Four games delimit the claims. Game 1 excludes accepted predecessor, step, or ADWC violations. Game 2 excludes a forwarded payload whose primary differs from the coordinate-derived primary. Game 3 hides coordinates and primaries between executions with equal L_S. Game 4 gives pure $\varepsilon_{\mathrm{total}}$-DP for the clipped released heatmaps under user-level adjacency, not for protocol metadata. Section V gives the bounds.

Admission functionality. The admission functionality $\mathcal{F}_{\text{SHTPC-Adm}}$ stores $(c_t, t, \mathrm{anchor}, \mathrm{warmup})$ per (uid, e) and forwards a report only when SHTPC, SRV-TS, SCS, and transport authentication hold. Warmup updates state without contributing a payload; reset starts a new epoch. The Shuffler receives L_S, each Aggregator receives one channel-local package, and the Decoder receives only aggregates. This is not a UC model of the full deployment.

Enrollment and Sybil resistance are deployment assumptions: each enrollment tuple must have threshold-valid EA attestations, and simultaneous Sybils remain bounded only by credential cost and external enrollment controls.

D. Violation Families and Coverage

The compact matrix below summarizes the concrete SHTPC-violation families used throughout the paper and separates circuit soundness from deployment-scope controls.

Family	Broken invariant	Coverage
A1 teleportation	Step continuity	Theorem 1
A2a substitution	Enrolled secret	Lemma 2
A2b transfer	Possession	Discussion
A3 gradual drift	K-window ADWC	Theorem 1
A4 Sybil creation	Enrollment	Proposition 2
A5 replay or reuse	Shuffler state	Theorem 1
A6 payload mismatch	Same primary	Theorem 2

E. Why Natural Compositions Are Insufficient

SHTPC is not obtained by placing a report-local proof before private aggregation. The following predicate-omission result collects the four relevant separations; it is not an impossibility theorem.

Proposition 1 (Composition-gap separations)**.** *An admission relation does not imply SHTPC if it omits any of the following: (i) the accepted predecessor, because individually valid states need not form a chain; (ii) the K-anchor predicate, because step-valid motion can drift beyond the window cap; (iii) a shared hidden primary, because trajectory and payload statements can open to different cells; or (iv) the hidden state-transition predicates, because context validation alone checks frame consistency rather than trajectory consistency.*

Proof. For (i), choose two locally valid states whose commitments are unrelated to the accepted predecessor. For (ii), let $x_i = x_0 + i\tau\Delta t\, e$; each step is valid while $\|x_K - x_0\| = K\tau\Delta t$ can exceed the cap. For (iii), prove for $p = \mathrm{Grid}(x)$ and route a payload committed to $p' \neq p$. For (iv), keep epoch, task, parameters, and blob hashes valid while choosing a state transition that the omitted hidden predicates would reject. □

These separations motivate a single accepted relation that binds predecessor state, the K-window anchor, the hidden coordinate, the coordinate-derived primary, and the routed payload. They also guide the evaluation design. We measure the SHTPC row, mechanism ablations, and the strongest low-cost stateful context-commitment harness. TEE and VDAF variants are treated as analytical baselines because they either change the privacy boundary or become equivalent to adding an SHTPC-style in-relation primary binding.

IV. Method

SHTPC inserts proof-carrying Shuffler admission into the ESA pipeline before secure aggregation. Its Client, EA, Shuffler, non-colluding Aggregators A/R, and Decoder realize Definition 2 without replacing the downstream private aggregation or DP release.

A. System Overview

Figure 1 shows the full pipeline. Each report first passes Shuffler-side proof and state verification. Accepted reports then contribute additive shares to Aggregators A and R, and the Decoder releases their aggregate as a DP heatmap. The rest of this section details the commitment chain and payload binders (§IV-C), the distance-verification circuit (§IV-D), enrollment and warmup (§IV-E), and the differential privacy layer (§IV-F).

In each reporting round, a client encodes its observation into additive shares of a clipped one-hot heatmap contribution. The primary cell is derived from the hidden coordinate. The client updates its commitment chain and generates a compact SHTPCMain proof. The proof binds the current hidden location, predecessor location, anchor location, identity commitment, tier/mode signals, hidden primary commitment, report-level payload commitment, and A/R share-content commitments while proving ADWC with zero-knowledge privacy.

Additive secret sharing [82] between the two Aggregators ensures that neither observes plaintext contributions alone. The implementation uses seed-compressed fixed-layout shares with VDAF-style expansion checks for single-Aggregator index hiding. Commitments are hash-based salted commitments instantiated with Poseidon over BN254 for efficient zero-knowledge proof circuits.

The Shuffler maintains an append-only per-uid, per-epoch accepted-state log under SCS. Accepted reports update the chain only after proof verification, timing-token validation, predecessor matching, payload-binder validation, and transport-authentication checks pass. Enrollment and reset enter warmup, where the same circuit path advances state but accepted payloads are withheld from aggregation.

B. Threat Model

The threat model considers malicious clients that may deviate arbitrarily from the protocol and may coordinate with one another. They know all public parameters, including tier speed bounds, ADWC caps, the window parameter K, and the verification key. They do not know honest clients' device-held secrets or raw coordinates.

The adversary may attempt the SHTPC-violation families A1 to A6 defined in §III-D. The circuit and Shuffler state address A1, A2a, A3, A5, and A6. A4 relies on EA threshold attestation against Sybil collusion [83]. A2b and false enrollment genesis points are deployment-scope issues handled by external controls rather than by the SNARK relation.

The server side follows the separated-role design of Prochlo and Nebula [4, 10]. The Shuffler is honest-but-curious. It observes proofs, commitments, ciphertext metadata, and accept/reject outcomes, yet it never sees plaintext coordinates. Aggregators A and R are further assumed not to collude. This organizational separation is an explicit deployment assumption. It is not implied by regulatory compliance or by assigning different service names to the roles. These are claim boundaries rather than engineering conveniences. If the Shuffler forks or rolls back state, the envelope-admission theorem is not claimed without an external audit log or replicated state machine. If the Shuffler colludes with an Aggregator, Game 3 no longer gives the stated single-server input-hiding view. If both Aggregators collude, input privacy degrades to the public output-DP guarantee. If the timing service lets clients choose Δt, the motion-envelope statement is not claimed.

Timing follows a server-issued assumption called SRV-TS. Under SRV-TS, the Shuffler honestly issues and verifies fresh time-slot tokens for each (uid, epoch, w), so clients are unable to choose, replay, or inflate $\mathrm{step\_dt\_sq}$ and prove only against a Shuffler-issued value. The token binds the user, epoch, round, previous accepted server time, current coarse server slot, squared interval, and nonce. This binding avoids relying on an unpredictable packet receive timestamp during proof generation while still preventing clients from selecting their own Δt.

Assumption SCS. Shuffler State Consistency. For each (uid, epoch), the Shuffler maintains an append-only accepted-state log with a monotonic round counter. A report is accepted only through an atomic compare-and-swap transition from the current predecessor state to a unique successor state, and the accepted record is durably logged before the payload is forwarded. Crash recovery restores the maximum durable accepted round. Duplicate rounds, forked successors, stale

TABLE II

DESIGN-SPACE BASELINES FOR THE SHTPC ADMISSION BOUNDARY. THE TABLE SEPARATES INCOMPLETE BUT PLAUSIBLE COMPOSITIONS FROM ALTERNATIVES THAT CHANGE THE PRIVACY OR TRUST MODEL.

Design	Coord. hidden	Stateful	Payload bound	Primary substitution	Cost or limitation
Stateless ZK + same-primary check	Yes	No	Partial	Partial	It can bind one report's proof and payload, but it does not express accepted predecessor state, replay prevention, or K-anchor consistency.
Stateful context commitment	Yes	Yes	Partial	No	The frame and blob hash are bound, but trajectory and payload statements may use different private primary witnesses.
TEE or plaintext filter	No or partial	Yes	Yes	Yes	It checks the predicate by trusting a runtime or revealing coordinates or primary cells to a checker. This is a privacy-cost upper bound.
VDAF/Prio with visible primary	No primary hiding	Partial	Yes	Yes	Validation can bind the encoded primary, but the primary is visible to the validation path.
VDAF/Prio with hidden unbound primary	Yes	Partial	No	No	Hidden measurement validation does not force the payload primary to equal the trajectory-derived primary.
VDAF/Prio with relation-bound primary	Yes	Depends	Yes	Yes	Preventing substitution requires an equivalent in-relation primary binding, which is the SHTPC subrelation studied here.
SHTPC	Yes	Yes	Yes	Yes	It binds accepted predecessor state, K-anchor reachability, coordinate-derived primary, and routed payload under explicit L_S.

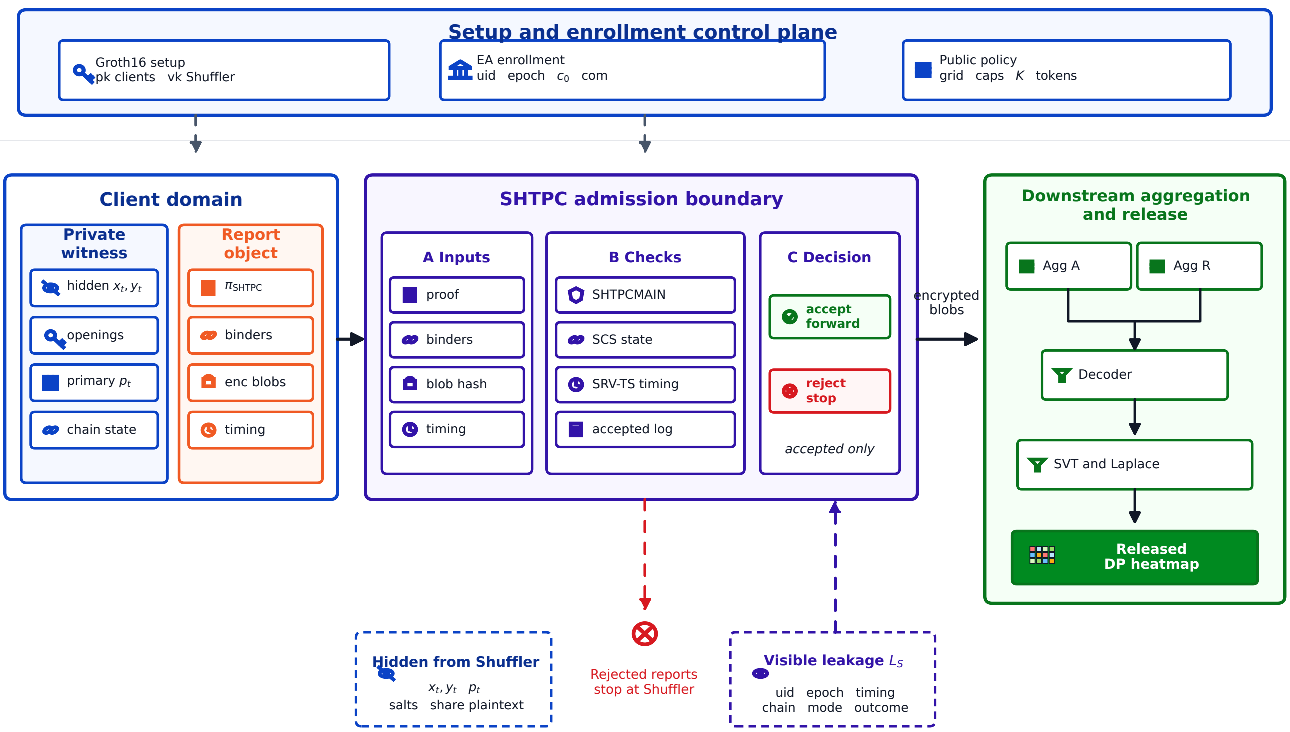

Fig. 1. Overview of SHTPC. Setup, enrollment, and public policy define the control plane. Each client report carries a proof, public binders, and encrypted payload blobs. The Shuffler enforces the admission boundary and forwards only accepted encrypted blobs to secure aggregation and downstream DP release. The lower callouts separate hidden witness values from Shuffler-visible leakage L_S.

predecessors, and rollback attempts are rejected. Reset creates a new epoch and re-enters warmup. The admission soundness theorem relies on SCS and does not cover a malicious Shuffler that violates this state discipline.

Enrollment authorization follows assumption CA3. Under CA3, at least t out of n enrollment anchors are non-colluding. The default profile is $t = 3$, $n = 5$, and a single-operator deployment is the degenerate case $t = 1$, $n = 1$.

Ground-truth enrollment and voluntary credential/device transfer (A2b) remain deployment risks outside Theorems 1–2. A transferred identity still obeys the envelope, timing, chain, and payload predicates; warmup requires $N_{\text{warm}} = \max(K, 2)$ plausible rounds, while Proposition 2 bounds only the cost of new identities. Deployments therefore need enrollment evidence, device binding, re-enrollment monitoring, distinct administrative domains, and audited cross-role flows. These controls make non-collusion auditable but do not enforce it cryptographically.

The component leakage boundary is summarized in Table I. The Shuffler sees SHTPCMAIN, public payload binders, blob_hash, fixed-size ciphertext metadata, timing tokens, and uid/epoch metadata, but not share plaintext, primary cells, or raw coordinates. Each Aggregator sees only its own decrypted share blob, round context, and validation outcome. The implementation invariants are fixed ciphertext length, fixed-layout expanded shares after local decoding, no primary-index logs, and Shuffler routing without share decryption.

C. Commitment Chain and Payload Binders

The commitment chain and payload binders provide the state-and-payload substrate needed by SHTPC. The chain makes the predecessor relation server-checkable without revealing coordinates, while the binders give the circuit a common handle for connecting the hidden primary cell to the routed share material.

Each client maintains a chain $\{c_w\}_{w=0}^{T}$. The per-round location commitment is $\ell_w = \mathrm{Poseidon}(x_w, y_w, r_w)$. Its salt is a circuit witness and is never exposed as a public input.

For compact notation, this paper writes $\mathsf{Com}_{\mathsf{pri}}$ for the hidden primary commitment, $\mathsf{Com}_{\mathsf{pay}}$ for the report-level payload commitment, and $\mathsf{Com}_A, \mathsf{Com}_R$ for the routed A/R share-content commitments. These binders are public inputs to SHTPCMAIN. Their openings are private witnesses.

The chain commitment is

$$\begin{aligned} c_w = \ &\mathrm{MiMC7}(c_{w-1}, \ell_w, t_w, w, \mathsf{Com}_{\mathsf{pri}}, \\ &\mathsf{Com}_{\mathsf{pay}}, \mathsf{Com}_A, \mathsf{Com}_R, \\ &\mathrm{com}_{\mathrm{sec}}), \end{aligned} \tag{3}$$

where c_{w-1} is the previous chain value, t_w is the Shuffler-issued timestamp slot, w is the round index, and the remaining terms are the public payload and identity binders for the same report.

The circuit derives the primary cell from the hidden coordinate exactly as in §III ($\mathrm{cell}_x = \lfloor x_w/s \rfloor$, $\mathrm{cell}_y = \lfloor y_w/s \rfloor$, and $\mathrm{primary} = \mathrm{cell}_y \cdot W + \mathrm{cell}_x$ with $s = 100\,\mathrm{m}$, $W = 100$, and $H = 100$). The primary cell and share-content digests are circuit witnesses, not public inputs. The circuit commits the hidden primary and binds the same committed primary to the report-level payload commitment and to the A/R share-content commitments under the current report context. The full binding relation (C21) is specified in Section IV-H. The key property is that all payload binders are derived from the same hidden primary and report context inside a single SHTPCMAIN statement.

The client sends seed-compressed fixed-layout share material with VDAF-style checks to Aggregator A and Aggregator R through separate ciphertext blobs. It publishes $\mathrm{blob\_hash} = H(\mathrm{blob}_A, \mathrm{blob}_R, \mathrm{ctx})$ for transport authentication. The hash H is collision resistant, and ctx is the report context. The Shuffler recomputes and verifies this digest against the proof public input to prevent post-proof ciphertext-package substitution. On receipt, each Aggregator decrypts its own blob and validates the encoded share package. The checks cover length, domain separator, context fields, and deterministic expansion to a D-cell layout. Aggregator-side validation is a format and context check. The hidden-primary semantic binding is enforced inside the circuit by SHTPCMAIN (C21).

Share representation and frame size. The privacy claim refers to the expanded aggregation view rather than to an uncompressed wire vector. After local expansion, every cell position has a share, and any one Aggregator's per-cell shares are indistinguishable from random without the other Aggregator's share. On the wire, the prototype sends seed-compressed encrypted share envelopes plus public proof material rather than a D-element wire vector.

Report frame and payload validation syntax. Each submitted report is represented as

$$\mathsf{Frame}_t = (uid, e, t, \pi_t, \mathsf{pub}_t, \mathsf{blob}_A, \mathsf{blob}_R, \sigma_t), \tag{4}$$

where uid, e, and t identify the pseudonymous admission chain, epoch, and round, π_t is the SHTPCMAIN proof, blob_A and blob_R are the encrypted Aggregator packages, and σ_t is the server-issued timing token and transport authentication material. The proof public statement contains

$$\begin{aligned} \mathsf{pub}_t = (&\mathsf{policy}, e, t, c_{t-1}, c_a, \mathsf{Com}_{\mathsf{pri}}, \\ &\mathsf{Com}_{\mathsf{pay}}, \mathsf{Com}_A, \mathsf{Com}_R, \mathrm{blob\_hash}, \mathrm{com}_{\mathrm{sec}}). \end{aligned} \tag{5}$$

The private witness contains the hidden coordinate, predecessor and anchor openings, the derived primary, commitment randomness, the clipped one-hot payload value, and the canonical share-content digests (d_A, d_R). The client computes

$$\mathrm{blob\_hash} = H(\mathsf{blob}_A, \mathsf{blob}_R, \mathsf{ctx}_t)$$

and opens Com_A and Com_R to the same $(\mathrm{primary}, d_A, d_R, \mathsf{ctx}_t)$ values used by C21. Each Aggregator decrypts only its own blob and accepts it only when the domain separator, context fields, channel label, length, deterministic expansion, and share-content digest match the proof-bound public binder. Thus the Shuffler checks proof and transport consistency, while Aggregators check channel-local package consistency. Neither role learns the hidden primary.

The identity commitment is

$$\mathrm{com}_{\mathrm{sec}} = \mathrm{Poseidon}(\mathrm{secret}, uid), \tag{6}$$

where $\mathrm{com}_{\mathrm{sec}}$ is the public identity commitment, secret is the client secret sampled at enrollment, and uid is the registered user identifier. The secret remains client-side. The off-chain tag

$$\mathrm{tag}_{\mathrm{sec}} = \mathrm{HMAC\text{-}SHA256}(\mathrm{secret}, uid) \tag{7}$$

where $\mathrm{tag}_{\mathrm{sec}}$ is an off-chain authentication tag. The tag is used only for enrollment and committee checks [84].

The Shuffler checks continuity by matching the next predecessor hash to the previously accepted current hash. The client therefore stores the salts needed to open the predecessor and the K-round anchor. For $K = 30$, the salt ring buffer costs $(K + 1) \cdot 32\,\mathrm{B} = 992\,\mathrm{B}$.

D. Distance Verification Circuit for Zero-Knowledge Proofs

The distance-verification circuit is the point where SHTPC's predicates become one accepted statement. It is designed to prevent a client from proving trajectory admissibility for one hidden primary while routing an aggregation payload for another, and to make all integer motion checks explicit inside the BN254 field.

The main circuit is parameterized by a compile-time window constant K. The evaluation instantiates two representative values, each with its own proving and verification parameters.

The Shuffler issues a signed timing token that binds $\Delta t = t_{\text{slot}} - t_{\text{last}}(uid)$ to (uid, epoch, w). Clients prove against the Shuffler-issued $\text{step\_dt\_sq} = \Delta t^2$ and are unable to choose or inflate their own Δt. The evaluated circuit is a fixed-slot artifact. Accepted non-gap reports use the configured $\Delta t_{\text{slot}} = 60\,\text{s}$, and session gaps re-enter warmup instead of enlarging the anchor window. A variable-slot deployment must replace the fixed-slot C14 derivation with a Shuffler-signed window-level cap. The defaults are $\Delta t_{\min} = 1\,\text{s}$ and $G = 30\,\text{min}$.

The public statement contains predecessor, current, and anchor location hashes, the squared step duration, tier and mode speed caps, anchor caps, $\mathsf{Com}_{\mathsf{pri}}$, $\mathsf{Com}_{\mathsf{pay}}$, Com_A, Com_R, blob_hash, secret commitment, modeset commitment, and the public mode tag. The witness contains the hidden coordinates, salts, primary-cell opening, share-content digest openings used by the payload-binding relation, secret material, and hidden mode selector bits.

The circuit has six constraint groups. C1–C3 bind salted location hashes. C4–C5 enforce movement constraints for each step. C6–C7 bind payload and identity. C8–C10 enforce mode membership and tag consistency. C11–C16 enforce ADWC anchor caps and cap/tier consistency. C17–C20 derive the hidden primary cell from the current coordinate. C21 binds that primary to $\mathsf{Com}_{\mathsf{pri}}$, $\mathsf{Com}_{\mathsf{pay}}$, Com_A, and Com_R under the same report context.

Full formulas are given in Section IV-H. All integer comparisons use LessEqThan(60) gadgets with explicit Num2Bits range checks on coordinates, squared distances, and public caps. Section IV-H1 gives the exact semantics. The implementation uses arity-3 Poseidon checks, and the evaluated relation compiles to 3,547 measured R1CS constraints with 16 public inputs and 25 private inputs.

Constraint C21 is the key relation that separates SHTPC from composing a trajectory proof with a VDAF-style check. C17–C20 derive $\text{primary} = W \cdot \lfloor y/s \rfloor + \lfloor x/s \rfloor$ from the hidden coordinate. C21 then commits that same primary in $\mathsf{Com}_{\mathsf{pri}}$, $\mathsf{Com}_{\mathsf{pay}}$, Com_A, and Com_R under report context ctx, with d_A, d_R as canonical A/R share-content digests.

Because C21 shares a circuit with C4–C5 and C11–C16, the accepted trajectory proof is bound to share material for the same primary cell. An out-of-circuit composition cannot enforce this linkage if the two proofs use independently chosen hidden primaries. The full binding formulas appear in Section IV-H, and the ablation in Section VI-B shows that removing C21 drops A6 rejection to zero.

The circuit proves openings for the predecessor, current, and anchor hashes. The Shuffler state machine enforces

$$\text{hash}_{\text{prev}}^{(w+1)} = \text{hash}_{\text{curr}}^{(w)} \tag{8}$$

and strictly increasing round indices through c_w. The state machine therefore prevents re-salting an old coordinate from replaying an accepted chain state.

E. Enrollment and Warmup

SHTPC cannot certify that the first committed location is ground truth, so enrollment and warmup define where the cryptographic claim starts. The design uses the same circuit path during warmup to avoid a weaker bootstrap mode, while withholding payloads until the chain has accumulated enough plausible state.

SHTPC uses one circuit path for bootstrap and steady-state proofs. For round w, the anchor index is

$$\text{anchor\_idx}(w) = \max(0, w - K), \tag{9}$$

where $\text{anchor\_idx}(w)$ is the index of the anchor state, w is the current round, and K is the compile-time window length. When $w < K$, the anchor is the genesis commitment c_0. Anchor constraints are never skipped.

Warmup is fixed to $N_{\text{warm}} = \max(K, 2)$ for enrollment and re-enrollment. During warmup, submissions are verified and advance chain state, but are withheld from aggregation. A session break occurs when the Shuffler-issued timing interval exceeds threshold G, set to $30\,\text{min}$ by default. After a break, the client re-enters warmup. Mode-only and tier-changing re-enrollment rules are specified in Section IV-G.

F. Differential Privacy Layer

After Shuffler acceptance, the Decoder applies SVT/Laplace to the clipped count vector with sensitivity $\Delta f = 1$, as defined in §III-A. In the default $T = 10$ configuration, $\varepsilon_{\text{total}} = 1.0$ and $\varepsilon_{\text{round}} = 0.1$, split into $\varepsilon_{\text{SVT}} = 0.03$ and $\varepsilon_{\text{val}} = 0.07$.

For each round, the Decoder scans fixed grid cells in deterministic order, releases cells whose noisy counts clear the public threshold, and stops at the grid end or the public release cap. The released object is the threshold-filtered noisy count vector and released cell set, not the raw aggregate. Unspent validation budget remains unused, and protocol-control metadata is excluded from the ε-DP accounting.

G. Protocol Operations

The steady-state report path keeps admission, aggregation, and release separate. After Shuffler acceptance, a report is split into additive shares and forwarded to Aggregators A and R, then the Decoder reconstructs the aggregate and applies SVT/Laplace before releasing the DP heatmap.

Re-enrollment deliberately returns to warmup instead of inheriting full admission rights. A mode-only change rotates mode_tag and keeps $\text{modeset\_commitment}$, while a tier change may update tier_vmax_sq and recompute the modeset commitment.

Timing is server-issued to avoid client-selected motion budgets. The timing token binds $(uid, \text{epoch}, w, t_{\text{last}}, t_{\text{slot}}, \Delta t^2, \nu)$, and submission verification checks the signature, freshness, uid/epoch/round binding, nonce, and equality with the proof public input. Requests with $\Delta t < \Delta t_{\min}$ are rejected, while gaps above G force warmup re-entry. The client stores (ℓ_w, r_w) for accepted states so the next proof can open the predecessor and the $\max(0, w + 1 - K)$ anchor. Reset starts a new epoch and never reuses salts.

H. Circuit Constraint Details

The circuit enforces the following constraints. C1–C20 cover trajectory, state, identity, mode, ADWC, and hidden-cell encoding. C21 is the in-circuit payload-binding relation. Table III summarizes the constraint groups, and §IV-H1 states the integer semantics used by the comparisons.

TABLE III
COMPRESSED CIRCUIT CONSTRAINT SUMMARY.

Group	Purpose
C1–C3	Salted Poseidon commitments for predecessor, current, and anchor locations.
C4–C5, C11–C16	Per-step speed bounds, ADWC anchor caps, cap/tier consistency, and selected-mode bounds.
C6–C10	Hidden-primary commitment, identity binding, modeset membership, and mode-tag consistency.
C17–C20	Coordinate-to-primary derivation with quotient and remainder constraints for cell size s, width W, and height H.
C21	Same-primary binding from $\mathsf{Com}_{\mathsf{pri}}$ to $\mathsf{Com}_{\mathsf{pay}}$, Com_A, and Com_R under report context ctx.

The arity-3 Poseidon implementation with explicit integer checks compiles to 3,547 R1CS constraints in the evaluated circuit.

1) Integer Semantics and Range Constraints: The BN254 scalar field has no native integer ordering, so every compared value is range-checked before use. Coordinates are signed 24-bit integers, coordinate differences are signed 25-bit integers, squared distances and public caps are 50-bit integers, $\mathrm{step\_dt\_sq}$ is 24-bit, and $\mathrm{tier\_vmax\_sq}$ is 16-bit. These bounds keep all compared values below 2^{60}. The circuit uses $\mathrm{Num2Bits}$ for coordinates, differences, distances, caps, and C17–C20 quotient/remainder witnesses, then implements all inequalities with $\mathrm{LessEqThan}(60)$. C17–C20 enforce $x_2 = q_x s + r_x$, $y_2 = q_y s + r_y$, and $\mathrm{primary} = q_y W + q_x$ with $0 \le r_x, r_y < s$, $0 \le q_x < W$, and $0 \le q_y < H$. The evaluated grid uses $s = 100\,\mathrm{m}$, $W = 100$, and $H = 100$. These range checks prevent field wraparound in the distance and cap comparisons and make primary-cell derivation unique without exposing $\mathrm{primary}$ to the Shuffler.

2) Private-Payload-Binding Tests: The implementation additionally mutates each public binder, ciphertext blob, channel label, replayed package, and report-context field after a valid baseline submission. All negative cases are rejected deterministically by C17–C21, Shuffler $\mathrm{blob\_hash}$ recomputation, or Aggregator-side format and context validation.

V. SECURITY AND THEORETICAL ANALYSIS

The implementation evidence used for this analysis is the evaluated Groth16 relation with 3,547 constraints and a 16/25 public/private input split.

Theorem 1 covers A1, A3, and A5. Lemma 2 covers A2a. Theorem 2 covers A6. A4 is bounded by the deployment proposition, while A2b and false genesis remain deployment-scope limitations.

Theorem 1 (Game 1 Envelope Admission Soundness)**.** *For any PPT adversary $\mathcal{A}$ making at most $q = \mathrm{poly}(\lambda)$ submission queries, if a submission is accepted by the* SHTPC *verifier and Shuffler state machine, and assuming Groth16 knowledge soundness, collision resistance of the hash relations inside the circuit, ciphertext-package authenticity, SRV-TS, and Shuffler State Consistency (SCS), then except with* $\mathrm{negl}(\lambda)$ *probability the extracted witness satisfies the following properties.*

E1 Continuity constraints for each step (C4/C5). *Under SRV-TS,*

$$\|p_i - p_{i-1}\|^2 \le \min(\mathrm{tier\_vmax\_sq}, \mathrm{mode\_vmax\_sq}) \cdot \mathrm{step\_dt\_sq}.$$

E2 Chain continuity. *The predecessor hash matches the previous accepted Shuffler state.*

E3 Anchor Distance Window Continuity (C11–C16).

$$\|p_i - p_{\max(0,i-K)}\|^2 \le \min(\mathrm{tier\_anchor\_cap\_sq}, \mathrm{cap\_policy\_sq})$$

and

$$\mathrm{cap\_policy\_sq} \le \mathrm{tier\_anchor\_cap\_sq}.$$

For q submissions,

$$\begin{aligned}\mathsf{Adv}^{\mathsf{traj}}_{\mathcal{A}} \le\ & q \cdot \mathsf{Adv}^{\mathsf{KS\text{-}Groth16}} \\ & + c_{\mathsf{P}} q \cdot \mathsf{Adv}^{\mathsf{CR\text{-}Poseidon}} \\ & + q \cdot \mathsf{Adv}^{\mathsf{CR\text{-}MiMC7}} \\ & + \mathrm{negl}(\lambda).\end{aligned} \tag{10}$$

Here c_{P} is the constant number of Poseidon commitment openings checked by SHTPCMAIN *per accepted submission.*

Proof. Let Win be the event that $\mathcal{A}$ outputs an accepted submission whose extracted witness violates at least one of E1–E3. The proof uses a sequence of games. In G_0, the challenger runs the real SHTPC acceptance experiment, so

$$\Pr[\mathsf{Win}_0] = \mathsf{Adv}^{\mathsf{traj}}_{\mathcal{A}}.$$

In G_1, the challenger aborts if the Groth16 knowledge-soundness extractor fails on any of the at most q accepting SHTPCMAIN proofs. By knowledge soundness and a union bound,

$$|\Pr[\mathsf{Win}_1] - \Pr[\mathsf{Win}_0]| \le q \cdot \mathsf{Adv}^{\mathsf{KS\text{-}Groth16}}.$$

In G_1, every accepted proof has an extracted witness satisfying the circuit relation. In G_2, the challenger aborts if any two distinct Poseidon openings collide across the location commitments and the commitment relations used by the continuity and payload-binding checks. Since SHTPCMAIN checks a constant number c_{P} of Poseidon commitment openings per accepted submission,

$$|\Pr[\mathsf{Win}_2] - \Pr[\mathsf{Win}_1]| \le c_{\mathsf{P}} q \cdot \mathsf{Adv}^{\mathsf{CR\text{-}Poseidon}}.$$

In G_3, the challenger aborts if two distinct Shuffler chain-state preimages collide under MiMC7. Collision resistance gives

$$|\Pr[\mathsf{Win}_3] - \Pr[\mathsf{Win}_2]| \le q \cdot \mathsf{Adv}^{\mathsf{CR\text{-}MiMC7}}.$$

In G_3, each accepted submission has a unique extracted witness satisfying the circuit relation and a unique chain-state preimage. Therefore C4/C5 and SRV-TS imply E1, the Shuffler state machine under SCS implies E2, and the ADWC constraints C11–C16 imply E3. Thus $\Pr[\mathsf{Win}_3] = 0$. Summing the three hop bounds yields Eq. (10). ☐

Lemma 1 (Same-Primary Binding)**.** *Let π be an accepted* SHTPCMAIN *proof with public binders* $(\mathsf{Com}_{\mathsf{pri}}, \mathsf{Com}_{\mathsf{pay}}, \mathsf{Com}_A, \mathsf{Com}_R)$ *and report context* ctx. *Under Groth16 knowledge soundness and Poseidon collision resistance, the extracted witness contains a unique hidden coordinate (x, y) such that* $\mathrm{primary} = W\lfloor y/s \rfloor + \lfloor x/s \rfloor$ *is the same value committed in all four binders, except with negligible probability.*

Proof. C17–C20 constrain the witness primary to equal $W\lfloor y/s \rfloor + \lfloor x/s \rfloor$ derived from the hidden coordinate (x, y). C6 binds this primary to $\mathsf{Com}_{\mathsf{pri}}$. C21 binds the same $\mathsf{Com}_{\mathsf{pri}}$ to $\mathsf{Com}_{\mathsf{pay}}$, Com_A, and Com_R under ctx. Under Groth16 knowledge soundness, extraction succeeds except with $q \cdot \mathsf{Adv}^{\mathsf{KS\text{-}Groth16}}$. Under Poseidon collision resistance, no adversary opens any binder to a different primary with non-negligible probability. Therefore all four binders identify the same hidden primary cell except with negligible probability. ☐

Theorem 2 (Game 2 End-to-End Payload Consistency)**.** *Assume Groth16 knowledge soundness for $\mathcal{R}_{\mathrm{SHTPC}}$, binding or collision resistance for the commitment and digest functions, ciphertext-package authenticity, deterministic share expansion, Shuffler* blob_hash *recomputation, and Aggregator-side format and context validation. For any PPT adversary that outputs an accepted and forwarded report frame* Frame_t, *except with negligible probability the decrypted Aggregator share packages decode to a clipped one-hot contribution whose primary cell equals the primary cell derived inside $\mathcal{R}_{\mathrm{SHTPC}}$ from the hidden coordinate witness. The theorem proves internal proof-to-payload consistency only. It does not prove that the coordinate is a true physical location.*

Proof. First, acceptance of π_t implies, by Groth16 knowledge soundness, an extracted witness containing $(x_t, y_t, \mathrm{primary}_t, \ldots)$ that satisfies $\mathcal{R}_{\mathrm{SHTPC}}$, except with negligible probability. Constraints C17–C20 force $\mathrm{primary}_t = W\lfloor y_t/s \rfloor + \lfloor x_t/s \rfloor$. Lemma 1 then gives the same $\mathrm{primary}_t$ across $\mathsf{Com}_{\mathsf{pri}}$, $\mathsf{Com}_{\mathsf{pay}}$, Com_A, and Com_R.

Second, the public binders and blob_hash bind the extracted primary to the report payload and the two channel-local share-content digests. Opening the same binder to an alternative primary $p'_t \neq \mathrm{primary}_t$ or replacing a proof-bound ciphertext package requires either a commitment or digest collision or a failure of ciphertext-package authenticity.

Third, each Aggregator accepts only if its decrypted package matches the proof-bound context, channel label, deterministic expansion rule, and share-content digest. Therefore the only share packages that can reach the Decoder are the packages bound to $\mathrm{primary}_t$ in the accepted proof statement. A successful proof-to-payload substitution would need to violate at least one of the assumptions above. Hence the decoded clipped one-hot contribution is internally consistent with the hidden coordinate-derived primary, except with negligible probability. ☐

Lemma 2 (Secret Binding)**.** *Under the assumptions of Theorem 1, any accepted submission opens the registered per-user secret commitment checked by C7, except with negligible probability.*

Proof. C7 enforces $\mathrm{Poseidon}(\mathrm{secret}, \mathrm{uid}_{\mathrm{field}}) = \mathrm{secret\_commitment}$ as a circuit constraint. Under Groth16 knowledge soundness, every accepted proof extracts a witness satisfying C7. Under the collision resistance of Poseidon, the extracted $(\mathrm{secret}, \mathrm{uid}_{\mathrm{field}})$ pair is unique to the registered secret_commitment. The EA registry binds each secret_commitment to exactly one enrollment identity. The Shuffler state machine enforces chain continuity per uid. Therefore an adversary who does not know secret cannot produce an accepting proof under a different uid's commitment. The advantage is bounded by $q \cdot \mathsf{Adv}^{\mathsf{KS\text{-}Groth16}} + q \cdot \mathsf{Adv}^{\mathsf{CR\text{-}Poseidon}} + \mathrm{negl}(\lambda)$. ☐

Payload-binding boundary. A successful A6 substitution across the gap between proof and payload requires one of four failures. The adversary must break the extracted payload-binding relation inside the circuit, produce a commitment collision, bypass Shuffler blob_hash checking, or make Aggregator-side validation accept malformed share material.

Tier and Sybil soundness are composed from circuit binding to the public tier/modeset inputs, registry matching against threshold-valid EA attestations, Ed25519 Existential Unforgeability under Chosen-Message Attack (EUF-CMA), CA3, and per-anchor credential cost.

Proposition 2 (Tier-Attestation and Sybil-Cost Deployment Bound)**.** *Any accepted tier must match a threshold-valid EA attestation. If each Sybil identity at tier τ requires t non-colluding anchors with expected per-anchor cost C_{anchor} and credential cost $c_{\mathrm{VC}}(\tau)$, then under budget B,*

$$m \leq \left\lfloor \frac{B}{t \cdot C_{\mathrm{anchor}} + c_{\mathrm{VC}}(\tau)} \right\rfloor . \tag{11}$$

This is a deployment-cost upper bound under the stated attestation and budgeting assumptions, not a cryptographic soundness theorem.

Theorem 3 (Game 3 Coordinate-and-Primary Hiding)**.** *Consider two executions that have identical public parameters, role corruptions, uid/epoch topology, timing bins, message-size classes, accept/reject traces, and all other Shuffler-visible leakage L_S, but may differ in hidden coordinates and primary cells. Against an honest-but-curious Shuffler or any one corrupted Aggregator, the distinguishing advantage for the hidden coordinates and primary cells is bounded by*

$$\begin{aligned} \Delta_{\mathsf{priv}} \leq\ & n \cdot \mathsf{Adv}^{\mathsf{ZK\text{-}Groth16}} \\ & + c_{\mathsf{H}} n \cdot \mathsf{Adv}^{\mathsf{HIDE\text{-}Poseidon}} \\ & + n \cdot \mathsf{Adv}^{\mathsf{IND\text{-}CPA/AEAD}} \\ & + n \cdot \mathsf{Adv}^{\mathsf{IndexHide\text{-}Share}} \\ & + \mathrm{negl}(\lambda), \end{aligned} \tag{12}$$

where n is the number of reports in the compared executions and c_{H} is the constant number of salted Poseidon commitments whose messages are hidden per report.

Proof. The proof is a standard hybrid over the corrupted server view. First, simulate SHTPCMAIN Groth16 proofs from the public statement. This simulation covers the entire circuit witness, including the trajectory witness, hidden-cell witness, and payload-binding witness. Second, replace the three location hashes and the public payload binders by salted Poseidon commitments to random messages. Third, replace Shuffler-routed ciphertexts with AEAD encryptions of dummy fixed-layout blobs. Finally, simulate any single corrupted Aggregator plaintext-share view by single-Aggregator index hiding. EA receives no trajectory data. The four hybrid hops give Eq. (12). □

The SHTPC-violation mapping in the opening paragraph identifies which formal statement or deployment assumption covers each SHTPC-violation family.

Game 3 protects coordinates and the primary cell under the fixed-leakage scope in §III-C. A corrupted Shuffler observing per-uid accept/reject sequences can detect coarse mobility patterns, including sleep schedules, mode changes, and commute regularity. This leakage is discussed in §VII. It does not reveal coordinate values or primary cells. Each result bounds one component of the adversary's advantage. Theorem 1 bounds acceptance of SHTPC trajectory-envelope violations. Theorem 2 bounds mismatches between proof and payload. Theorem 3 bounds coordinate-and-primary distinguishability under equal L_S. Game 4 covers the released heatmap, while enrollment is handled by the deployment proposition. A joint adversary's advantage is therefore bounded by the sum of these game advantages plus the leakage excluded from $\varepsilon_{\text{total}}$.

VI. ADMISSION-BOUNDARY ENFORCEMENT EVALUATION

The evaluation follows the narrowed paper claims. It does not measure general location truthfulness or poisoning robustness. RQ1 tests whether SHTPC rejects proof-to-payload inconsistent reports. RQ2 tests whether stateful predecessor and K-window checks enforce the public envelope. RQ3 quantifies what remains inferable from Shuffler-visible metadata L_S. RQ4 characterizes residual in-envelope and deployment-scope risks. RQ5 measures heatmap utility after admission and DP release. RQ6 measures proof-carrying execution cost and scalability. All rejection results should be read as boundary-enforcement results.

A. Experimental Setup

Datasets and evaluation tracks. The four real inputs are T-Drive (10,357 taxis, about 15M points), GeoLife (182 users, over 17K trajectories), Porto, and Rome; the fifth is a clustered random-walk Synthetic input (seed 20260709) [1, 2, 85–89]. Each prepared input contains 1,000 records and ten windows without bootstrap resampling. The evaluation is organized by claim rather than by dataset alone: RQ1–RQ2 test mechanism removal and state/window boundaries, RQ3 measures metadata leakage, RQ4 stresses residual attacks, RQ5 compares released-heatmap utility, and RQ6 measures full proving scale. Dense T-Drive and GeoLife subsets additionally stress rush-hour mobility and session gaps.

Baselines and fairness. Admission experiments compare SHTPC-FULL with four measured variants: NO-ADMISSION-CHECK, NO-ADWC, NO-PAYLOAD-BIND, and STATEFUL-CONTEXTCOMMIT. The last is the strongest coordinate-hiding composition baseline: it binds identity, epoch, round, predecessor, frame, and share context, but proves no same-primary equality. Table II maps these variants to their missing predicates. A trusted plaintext filter is only an analytical upper bound because it reveals coordinates. For RQ5, a generic external ESA (Nebula-style) adapter and an EIFFeL-style range/norm adapter run on the same reports. The ESA adapter is the strongest external utility comparator and is shown in Figure 5; the EIFFeL-style adapter has no cross-round rule and rejects none of the valid-range A1 reports, so it is not presented as a privacy-equivalent release mechanism. These are mechanism-level adapters, not reproductions of complete upstream deployments. No baseline is tuned per dataset, and all fixed-utility methods use the same inputs, attacker assignment, seeds, and top-50 Jaccard metric.

Attacks, parameters, and metrics. The controlled suite covers A1 teleportation, A2a secret substitution, A3 drift, A5 replay/state reuse, and A6 proof-payload substitution; A4 and A2b are deployment-scope cases. Controlled runs use $N = 50, T = 10$; utility uses $N = 1000, T = 10$. Unless stated otherwise, $\varepsilon_{\text{total}} = 1$, $\Delta t = 60\,\text{s}$, $K \in \{6, 30\}$, 100-m cells over a 10-km domain, and walk/bike/vehicle/transit caps of 3/10/33/40 m/s. The policy caps are 3.3 km ($K = 6$) and 16.5 km ($K = 30$); the latter is a stress profile. Reports are L_1-clipped to one cell. TPR/FRR measure malicious/benign rejection; Jaccard measures released-set stability; RMSE, Pearson, false-hotspot probability, and HS@k measure count and ranking distortion. Randomized experiments use three seeds unless noted.

Implementation and execution protocols. The Python/Circom/snarkjs prototype ran in Docker on an Apple M4 MacBook Pro (10 CPU cores, 16 GB RAM; Docker: 10 CPUs, 8.22 GB) using Python 3.10.20, Node.js 20.20.2, Circom 2.1.9, snarkjs 0.7.6, and Docker Server 29.1.3 on Linux arm64. RQ5 reads the original inputs with seeds 101/202/303 and one fixed release configuration. Separately, five clean protocol-smoke runs replace gaps and out-of-envelope jumps with safe trajectories to isolate proving, state, reconstruction, and release; they are not utility evidence. RQ6 uses the same protocol-isolation trajectories for a 15-unit matrix (five datasets, three seeds, six warmup plus ten evaluation rounds). Functional totals retain every passing unit; timing excludes only units whose maximum inter-round gap exceeds 7,200 s, with at least two eligible units required per dataset.

B. RQ1 Proof-to-Payload Consistency

RQ1 asks whether SHTPC rejects reports whose proof and forwarded payload are inconsistent. The main target is A6, where a valid trajectory proof is paired with payload material for a different primary cell. We also report the other SHTPC violation families in the same matrix to show which mechanism guards each admission boundary. Figure 2

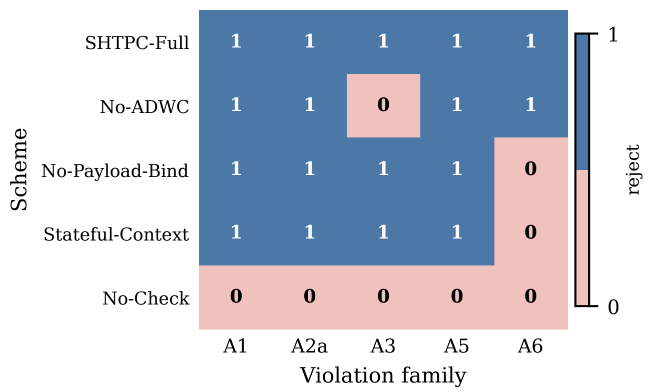

Fig. 2. Mechanism-level admission ablation on GeoLife. A value of 1 means rejection and 0 means admission. ADWC is necessary for A3, while same-primary payload binding is necessary for A6.

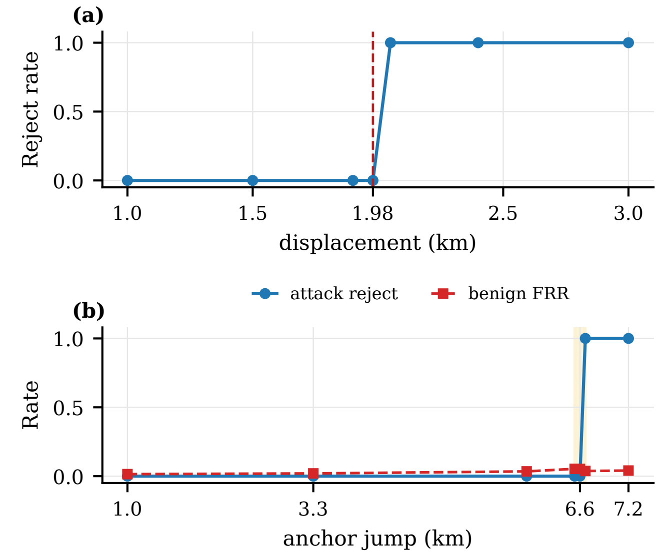

Fig. 3. Boundary sweeps for the two trajectory predicates. Panel (a) plots per-step rejection around the vehicle-tier displacement cap. Panel (b) plots attack rejection and benign FRR around the ADWC anchor boundary.

reports the GeoLife within-pipeline ablation under the default controlled configuration. Each row removes one mechanism and shows which violation family becomes admissible.

SHTPC-FULL rejects all modeled A1, A2a, A3, A5, and A6 violations, so Figure 2 is an admission-consistency result. Primary-cell hiding is a formal Game 3 claim under equal L_S, and post-admission utility is evaluated separately in RQ5. NO-ADMISSION-CHECK accepts every modeled violation. NO-ADWC loses A3 because gradual drift is the anchor-window invariant. NO-PAYLOAD-BIND and STATEFUL-CONTEXTCOMMIT both admit A6 because trajectory and payload statements are separated.

A6 is also tested through a private payload-binding harness with exactly 12 deterministic negative attempts against one passing fixture. Each mutation is attempted once. Two attempts substitute proof or public-signal context. Five substitute the hidden primary, A-share digest, R-share digest, both share digests, or the private context commitment. Five mutate the post-proof A share, R share, cross-routed shares, route context, or blob hash. All 12 attempts are rejected across proof, transport, and aggregation validation. The 22-check static gate also includes five independent state or identity mutations and five passing compatibility cases.

C. RQ2 Stateful Envelope Boundary

RQ2 asks how statefulness affects predecessor, replay, and K-window consistency. The measured STATEFUL-CONTEXTCOMMIT baseline represents a strong context-binding composition without in-relation primary equality. It rejects stale predecessor replay, gradual drift, timing-token replay, and context substitution, but it still admits A6 because the trajectory proof can be valid for one hidden primary while the payload is routed to another. SHTPC-FULL is the only coordinate-hiding row in Figure 2 that rejects A6 because the trajectory-derived primary and payload primary are constrained inside one relation.

Boundary sharpness. Boundary sweeps separate the two continuity predicates. For E1, vehicle-tier proof generation accepts displacements up to $\tau_{\text{vehicle}}\Delta t = 1980\,\text{m}$ and rejects larger displacements. For E3, the T-Drive ADWC sweep at $K{=}6$, vehicle tier, and $\text{cap}_{\text{policy}} = 3.3\,\text{km}$ transitions near $6.6\,\text{km}$ in the plotted attack-jump parameterization because the sweep moves the adversarial endpoint across both sides of the anchor boundary. The underlying admission predicate remains the $3.3\,\text{km}$ anchor-distance cap. The observed transition reflects the windowed anchor-distance boundary rather than a single-step speed cap.

A3 is evaluated as a *parameterized ADWC admission check*. Two deployment profiles are used, $K{=}6$ ($\text{cap}_{\text{policy}}{=}3.3\,\text{km}$) and $K{=}30$ ($\text{cap}_{\text{policy}}{=}16.5\,\text{km}$). For $K{=}6$, TPR= 1.00 across all five drift biases (0.20–1.00). For $K{=}30$, TPR ranges from 0.80 (bias= 0.20) to 0.94 (bias= 0.40), reflecting the larger window's softer resolution for each step. $K{=}30$ is a stress test rather than the recommended deployment. Tightening $\text{cap}_{\text{policy}}$ from 1.67× to 0.67× default raises $K{=}30$ TPR from 0.89 to 0.99 in the policy-cap sensitivity sweep.

(τ, τ_2) sensitivity and session-gap baseline. A (τ, τ_2) sweep over $\{1, \ldots, 5\}^2$ (75 runs, 3 seeds per pair) confirms DP-parameter invariance. Mean TPR 0.981 (range 0.951–1.000) and session-withholding rate exactly 0.192 across all pairs (circuit-FRR 0). This is expected because proof verification precedes the DP/SVT release stage.

D. RQ3 Shuffler-Visible Metadata Leakage

RQ3 asks what behavior remains inferable from L_S. The adversary observes only Shuffler-visible metadata such as chain length, timing, topology, and accept/reject histories. Every reported classifier fits a StandardScaler on the training set and then trains a two-hidden-layer MLP with 32 and 16 ReLU units, Adam, an L_2 coefficient of 10^{-4}, learning rate 10^{-3}, and 15% early-stopping validation. The iteration caps are 150 for the synthetic ablation and 500 for the real-trace tasks.

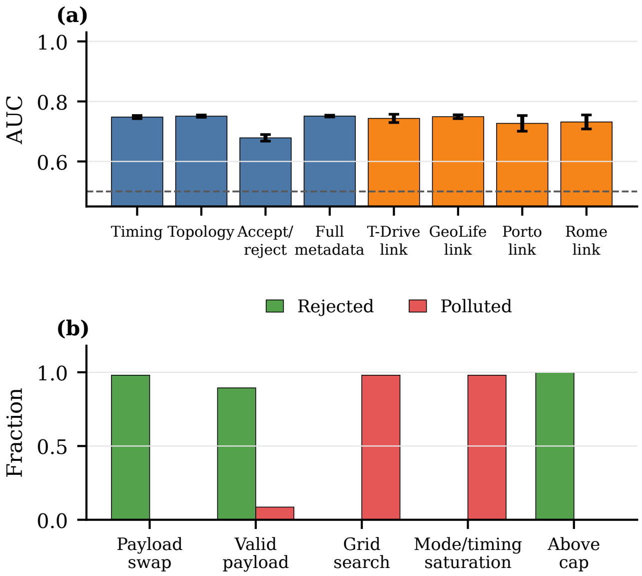

Fig. 4. Dense evaluation summary. Panel (a) reports synthetic feature-view ablations in blue and user-disjoint real-trace session linking in orange. The dashed line marks random AUC. Panel (b) reports adaptive boundary outcomes and separates rejected attempts from accepted polluted outcomes.

The synthetic ablation uses a label-stratified 70/30 sample split. Each seed has 8,750 training samples and 3,750 test samples. With 500 users, 30 rounds, delay mean 1.5 rounds, and seeds 7, 11, and 19, timing-only and chain-topology views reach AUC 0.748 ± 0.005 and 0.751 ± 0.004. Full metadata reaches 0.751 ± 0.003. Accept/reject history reaches 0.679 ± 0.011.

The real session-linking task uses a stricter user-disjoint split. Each trace is divided at its temporal midpoint. Every user contributes one same-user pair and one cross-user pair. Users are split 700/300 before pairs are generated, and negative partners are drawn only within the same partition. This produces 1,400 balanced training pairs and 600 balanced test pairs per seed with no user shared across partitions. At $N = 1000$, the 30-bin selection cap yields 17 T-Drive, 18 GeoLife, 11 Porto, and 8 Rome hourly bins. Mean AUC and sample standard deviation over seeds 7, 11, and 19 are 0.7434 ± 0.0138, 0.7491 ± 0.0064, 0.7268 ± 0.0261, and 0.7316 ± 0.0233, respectively. Session linking is therefore consistently above random on all four real inputs. Active-period AUC ranges from 0.9987 to 0.9999. Thus SHTPC hides coordinates and primary cells under Game 3, but not participation, chain growth, timing regularity, or accept/reject traces.

E. RQ4 Residual In-Envelope and Deployment-Scope Risks

RQ4 asks what remains after the cryptographic envelope is enforced. SHTPC bounds a malicious client's spatial influence per round to at most $\tau_u \Delta t$ and per K-window to $\mathrm{cap}_{\mathrm{policy}}$. Under the default vehicle profile, where $\tau_u \Delta t = 1980\,\mathrm{m}$, one accepted report can target at most 1,229 cells, or about 12.3% of the grid. Under K=6 and $\mathrm{cap}_{\mathrm{policy}} = 3.3\,\mathrm{km}$, the per-identity per-window reach is at most 3,409 cells, or 34.09% of the grid. This reach bound limits where accepted malicious mass can appear. It does not multiply down the malicious clients' total L_1 mass, because an adaptive adversary may concentrate its clipped contribution on one reachable cell.

Proposition 3 (Accepted Malicious Mass and Support Bounds)**.** *Let ρ be the malicious-client fraction in a reporting window. Let C_{clip} be the per-report L_1 clipping constant. Let $R_u \subseteq \mathcal{G}$ be the set of cells reachable by malicious client u under the public $\mathrm{cap}_{\mathrm{policy}}$, with $|R_u| \leq A_{\mathrm{cap}}$. For the normalized heatmap perturbation $\Delta\bar{h}$ caused by accepted malicious reports,*

$$\|\Delta\bar{h}\|_1 \leq \rho C_{\mathrm{clip}}, \qquad \|\Delta\bar{h}\|_\infty \leq \rho C_{\mathrm{clip}},$$

and

$$|\mathrm{supp}(\Delta\bar{h})| \leq \left| \bigcup_{u \in M} R_u \right| \leq \min(|\mathcal{G}|, |M| A_{\mathrm{cap}}).$$

Thus A_{cap} bounds the spatial support of accepted malicious influence, but it does not reduce the worst-case L_1 mass.

Proof. Each accepted malicious report has total count weight at most C_{clip}, which gives the normalized L_1 bound ρC_{clip}. All malicious clients may target the same reachable cell, so the same value is the worst-case per-cell bound. The support bound follows from taking the union of reachable sets R_u. $\square$

Joint adaptive adversary. SHTPC enforces an envelope, not ground truth. In the joint adaptive suite (N=100, 10% malicious clients, T=50, K=6, $\mathrm{cap}_{\mathrm{policy}}$=3.3 km), payload-decoupled attempts are rejected in 98% of malicious cell-rounds and admit no in-envelope contribution. Payload-bound in-envelope attempts are accepted by design (490/500 cell-rounds), while the same strategy at 1.01× the ADWC cap is rejected in all 500 malicious cell-rounds. False genesis and A2b credential/device transfer remain deployment-scope risks. With G=30 min, inter-window gaps exceeding G are 1.10% on T-Drive and 2.12% on GeoLife.

F. RQ5 Utility After Admission and DP Release

RQ5 asks how much utility is lost after adding SHTPC before the unchanged SVT/Laplace DP release. The headline protocol fixes ε=5.0 and the post-admission release threshold τ=2.0 for every dataset, uses N=1000, T=10, 10% A1 attackers, top-50 released-set Jaccard, and seeds 101, 202, and 303. It does not substitute per-dataset tuned settings. Among the evaluated fixed-configuration adapters, the external ESA adapter has the highest mean Jaccard on every dataset [10, 11].

Figure 5 reports mean Jaccard for SHTPC and the strongest external utility adapter. The respective values are 0.671/0.561 on T-Drive, 0.878/0.366 on GeoLife, 0.305/0.298 on Porto, 0.772/0.664 on Rome, and 0.135/0.128 on Synthetic. The absolute gains are therefore +0.110, +0.512, +0.007, +0.108, and +0.007, respectively. Porto and Synthetic provide only small gains, so the result supports an observed positive direction under the fixed protocol rather than a uniformly large margin. All 15 observed dataset-seed paired differences

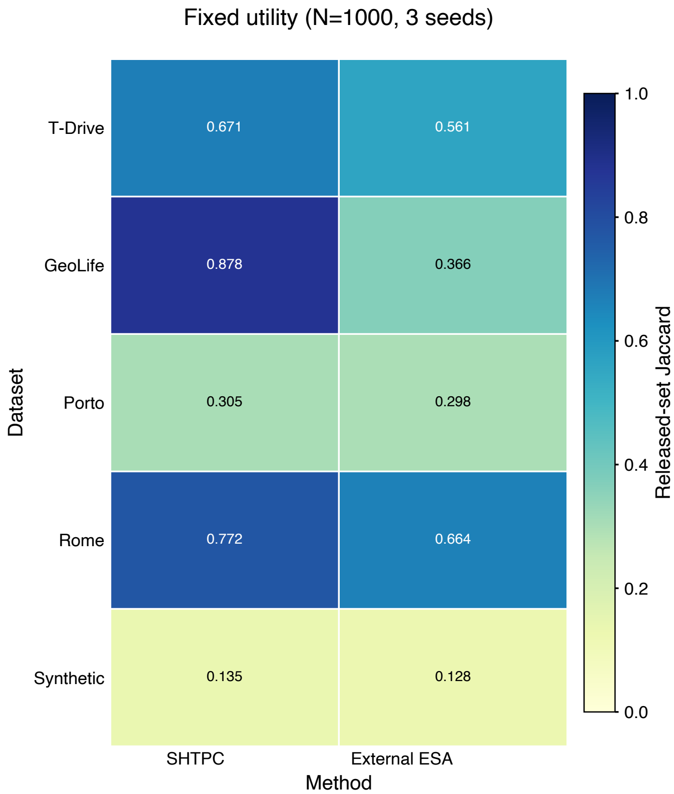

Fig. 5. Fixed-configuration released-set Jaccard at N=1000 and T=10, averaged over seeds 101, 202, and 303. Both methods use ε=5.0. SHTPC uses post-admission threshold 2. The generic external ESA adapter uses $p = 1-\exp(-\varepsilon)$: it emits the submitted cell with probability p, otherwise a uniform dummy over D = 10,000, and shuffles reports. It filters raw counts below 3; with $q = (1-p)/D$, it outputs $\max\{0, (c-Nq)/(p-q)\}$. Across datasets, SHTPC standard deviations are 0.006–0.044 and external-ESA standard deviations are 0.000–0.003. SHTPC has FRR 0 and MRR 1; both external adapters have MRR 0 for valid-range A1 reports.

favor SHTPC, and the smallest is +0.00239. Each dataset has only three seeds, so we do not claim statistical significance for the small Porto and Synthetic differences. SHTPC also has zero honest false rejection and complete modeled malicious rejection on every dataset in this comparison. A single-round EIFFeL-style range/norm adapter yields Jaccard 0.543/0.300/0.294/0.644/0.127 on the same dataset order and rejects none of the valid-range A1 reports; it is dominated by the external ESA adapter on utility and does not match SHTPC's stateful admission boundary. A separate GeoLife sensitivity sweep found admission TPR 1.00 at each tested ε, consistent with proof verification preceding the release stage. It is supporting evidence rather than the headline utility comparison. As shown in RQ4, accepted in-envelope mass can still create false hotspots, which remains a deployment risk.

G. RQ6 Cost and Scalability

RQ6 asks whether proof-carrying admission is practical at per-report granularity. The evaluated Groth16 relation has 3,547 constraints, 16 public inputs, and 25 private inputs. It passes all 22 static checks and clean protocol smoke on 5 datasets. The strict three-seed N=1000 aggregate verifies all 15 dataset-seed units, with 225,000 client proofs generated and

TABLE IV
GROTH16 PROTOCOL EXECUTION AT N=1000 OVER THREE SEEDS. ALL FUNCTIONALLY VALID UNITS CONTRIBUTE TO PROOF TOTALS. TIMING EXCLUDES UNITS WITH AN INTER-ROUND GAP ABOVE 7,200 SECONDS; ENTRIES REPORT MEAN $\pm$ SAMPLE STANDARD DEVIATION OVER THE REMAINING TIMING-ELIGIBLE UNITS.

Dataset	Verified	Timing n	Proofs/s	Wall time (h)
T-Drive	45,000	3	1.072 $\pm$ 0.127	3.921 $\pm$ 0.439
GeoLife	45,000	3	0.994 $\pm$ 0.027	4.195 $\pm$ 0.113
Porto	45,000	3	0.967 $\pm$ 0.020	4.308 $\pm$ 0.091
Rome	45,000	2	0.967 $\pm$ 0.031	4.310 $\pm$ 0.138
Synthetic	45,000	2	0.983 $\pm$ 0.009	4.240 $\pm$ 0.039

Shuffler-verified and 0 proof failures. Across 150 evaluation rounds, routes A and R receive 150,000 and 150,000 reports, respectively. All 150 reconstructions and 150 DP releases pass.

Table IV separates functional completion from timing eligibility. Rome seed 101 and Synthetic seed 101 each contain an inter-round gap above 62,962 seconds caused by host suspension. Their functional records remain in every proof, route, reconstruction, and release total, but the two units are excluded only from timing statistics under the 7,200-second rule. The remaining 13 units, with 2 exclusions, average 1.000 ± 0.069 proofs/s and 4.183 ± 0.248 hours per unit (mean $\pm$ sample standard deviation across timing-eligible units). Thus the prototype establishes reproducible full-matrix execution with no proof or release failure, but it does not yet provide interactive per-report latency or optimized production throughput.

VII. Discussion and Limitations

The TCB comprises the Groth16 setup, EA Ed25519 keys, authenticated channels, and Poseidon/MiMC7 assumptions over BN254 [90]. Aggregator collusion reduces input privacy to the public DP release; a compromised setup threatens soundness, and a malicious Shuffler requires replicated state or external audit. SHTPC is therefore intended for managed deployments with accountable enrollment and separated roles, not fully anonymous or unlinkable reporting.

Metadata remains an explicit boundary. The Shuffler observes L_S, including chain topology, timing, tier, message size, and outcomes; output DP does not protect these signals. Fixed-rate batching, dummy warmup traffic, coarser time bins, reset cooldowns, and auditable logs can reduce exposure, but change latency or bandwidth and must be included in L_S before extending Game 3.

SHTPC also enforces an envelope rather than ground truth. In-envelope false reports, false genesis, and credential/device transfer (A2b) remain possible. They require enrollment proof, device attestation or secure elements, quotas, behavioral controls, and re-enrollment monitoring rather than additional circuit predicates.

A. Artifact and Reproducibility Details

The artifact index binds source, R1CS/verification-key hashes, public-signal order, C21, grid/hash parameters, frame construction, share expansion, static checks, protocol smoke, fixed utility, and all 15 scale records. Real datasets must be obtained from their providers under `DATA_NOTICE.md`; raw and row-level prepared traces are not redistributed. Preparation code, seeds, aggregate receipts, and the fixed-seed Synthetic generator are included. The checked-in Groth16 material is for development reproducibility, not evidence of a production ceremony; any relation or parameter change requires regeneration and validation.

B. Ethics, Governance, and Misuse

The evaluation reuses provider-released traces and collects no new participant data. Reproducers must follow provider license, citation, redistribution, and non-de-anonymization terms. Because persistent admission state can reveal participation and behavioral patterns, deployments need explicit consent, retention, revocation, access-control, and audit policies. Logs should be minimized and time-limited, and high-risk uses require external oversight. Operators must state that SHTPC provides neither physical-presence proof nor trajectory anonymity.

VIII. Conclusion

This paper addresses hidden trajectory-payload mismatch in private heatmap aggregation through the SHTPC admission property. Its key idea is to bind predecessor continuity, K-window reachability, hidden-primary derivation, and payload equality inside one zero-knowledge relation before aggregation separates the report across roles. The evaluated Groth16 relation contains 3,547 constraints, 16 public inputs, and 25 private inputs. The evaluation shows that SHTPC rejects the modeled envelope and proof-payload violations admitted by its ablations and by Stateful-ContextCommit. The fixed comparison also yields positive observed Jaccard differences on all five datasets, although two differences are small. Across 15 $N{=}1000$ units, the full-path matrix verifies 225,000 proofs with no proof, reconstruction, or DP-release failure. Its roughly 4.183-hour mean unit runtime supports reproducible execution rather than interactive latency. These results support managed heatmap systems with stateful admission and explicit governance. They do not extend to physical-presence proof, metadata anonymity, fully in-envelope false reports, or a malicious Shuffler without external accountability.

Future Work. Future work should evaluate stronger enrollment, device attestation, auditable Shufflers, larger adaptive sweeps, and universal-setup proof-system ports such as PLONK [71].

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